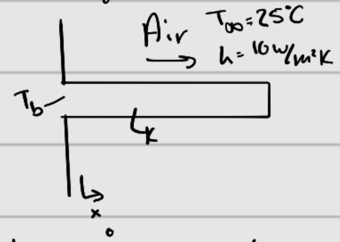
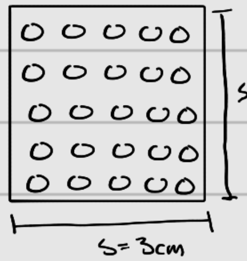
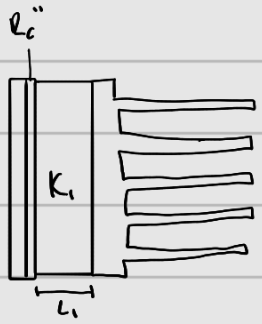
1. Known: $r = 1mm$ $L = 5cm$ $T_\infty = 25°C$ $h = 10\ W/m^2K$ $K = 200\ W/mK$

Find: a) Fin Temp @ $x = L/2$, $x = L$ ($T_b = 125°C$) b) $\varepsilon_f$ $\eta_f$ via Table 3.4 c) $\eta_f$ via T 3.5 compare to (b)
d) $R_f$ e) $\eta_o$ & $R_o$ f) $L_1 = 2cm$, $R_c = 10^{-4}\ m^2K/W$, $q'' = 1000\ W/m^2$, $K_1 = 10\ W/mK$ Thermal Resistance + $T_h$, Compare

Schematic:




Assumptions: Steady State, 1-D Conduction, Const. Prop., Uniform conv.
Neglect Radiation, Adiabatic Tip

Analysis:

$A_c = \pi r^2 = 3.14 \cdot 10^{-6}\ m^2$ $P = 2\pi r = 6.28 \cdot 10^{-3}\ m$ $m = \left(\frac{hP}{KA_c}\right)^{1/2} = 10\ m^{-1}$

$M = \sqrt{hPA_cK}\,\theta_b = 0.628$

a) $T(x) - T_\infty = (T_b - T_\infty)\left(\frac{\cosh m(L-x)}{\cosh mL}\right)$

$T_b - T_\infty = 100°C$

$T(L/2) = 100\left(\frac{\cosh 10(\frac{.05}{2})}{\cosh 10(.05)}\right) + 25°C = 116.47°C$

$T(L) = 100\left(\frac{\cosh 10(\emptyset)}{\cosh 10(.05)}\right) + 25°C = 113.68°C$

b) $q_f = M \tanh(mL) = .290W$ $A_f = PL = 6.28 \cdot 10^{-3} \cdot 0.05 = 3.14 \cdot 10^{-4}$

$\eta_f = \frac{q_f}{hA_f\theta_b} = .924$ $\varepsilon_f = \frac{q_f}{hA_c\theta_b} = 92.36$

c) $\eta_f = \frac{\tanh(mL_c)}{mL_c} = 0.923$ This is very close to (b). The slight diff is probably from approximating the tip

$L_c = .05 + \frac{0.001}{2} = 0.0505$

d) $R_f = \frac{T_b - T_\infty}{q_f} = 0.345\ K/W$ e) $s = .03m$ $5(5) = 25 = N$

$A_{uf} = (.03)^2 - 25(3.14 \cdot 10^{-6}) = 8.15 \cdot 10^{-4}\ m$

$A_{tot} = NA_f + A_{uf} = 8.665 \cdot 10^{-3}\ m$

$Q_{fins} = Nq_f = 7.25\ W$ $\eta_o = \frac{N\eta_f A_f + A_{uf}}{A_{tot}} = .930$

$Q_{uf} = hA_{uf}(T_b - T_\infty) = 0.815W$ $R_o = \frac{1}{\eta_o h A_{tot}} = 12.41\ K/W$

f) $L_1 = 2\,cm$ $\quad k_1 = 10\ W/mK$ $\quad R_c'' = 10^{-4}\ \frac{m^2K}{W}$ $\quad q'' = 1000\ W/m^2$

$$Q \rightarrow \underset{T_h}{\circ} \overset{R_s}{\text{—}\!\!\vee\!\!\vee\!\!\text{—}} \underset{T_I}{\circ} \overset{R_c}{\text{—}\!\!\vee\!\!\vee\!\!\text{—}} \underset{T_b}{\circ} \overset{R_{fin}}{\text{—}\!\!\vee\!\!\vee\!\!\text{—}} \underset{T_\infty}{\circ}$$

$Q = q''A = 0.9\,W$ $\qquad R_s = \frac{L_1}{kA} = 2.22\ K/W$ $\qquad R_c = \frac{R_c''}{A} = 0.111\ K/W$

$R_{fin} = \frac{T_b - T_\infty}{Q_f + Q_{uf}} = 12.4\ K/W$ $\qquad R_{tot} = R_c + R_s + R_{fin} = 14.7\ K/W$

$T_h = T_\infty + R_{tot}Q = 38.2^\circ C$

If there was no heat sink: $Q = hA(T_h - T_\infty)$

$T_h = \frac{Q}{hA} + T_\infty = 125^\circ C$

The heat sink reduces the temp by $87^\circ C$